NDINAYO ANICET

Master's Student in Mathematical Science (Epidemiology)

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EDUCATION

Master's in Mathematical Science (Epidemiology)

AIMS Rwanda

2025 – 2026

- Specializing in mathematical modeling for disease spread and public health
- Advanced coursework in statistical methods and computational mathematics

Bachelor's in Applied Statistics (Demography)

University of Rwanda

2022 – 2025

- Thesis: Rwanda Population Projection 2050 - Analyzing demographic trends and labor market impacts
- School of Business and Economics

WORK EXPERIENCE

Data Collection Cooperator

National Institute of Statistics of Rwanda (NISR)

2023

- Conducted field data collection for Demographic and Health Survey (DHS)
- Applied rigorous data validation and quality assurance protocols
- Collaborated with team to ensure timely project completion

TRAININGS & CERTIFICATIONS

- **Data Analysis by using Programming** - Python and R programming (University of Rwanda - 2024)
- **NISR Training Program** - Data Collection Methods (2023)
- **Statistical Software Certification** - Stata and Advanced Excel
- **Research Methodology Workshop** - University of Rwanda (2023)

PROJECTS

Optimization of Resource Allocation for Cholera Prevention in Burundi

Master's Thesis and Internship Project

May 2026–Sept 2026

PROFILE

A passionate researcher with strong interest in applying mathematical and statistical methods to solve public health challenges. Enthusiastic about epidemiological modeling, data driven decision making, and using computational tools to address real world problems. Committed to continuous learning and contributing to evidence based solutions in healthcare and development sectors.

TECHNICAL SKILLS

- **Statistical Analysis**
- **R Programming**
- **Python**
- **Stata**
- **Advanced Excel**
- **LaTeX**
- **Data Collection Tools**
- **Microsoft Office**

SOFT SKILLS

- **Problem Solving**
- **Communication**
- **Teamwork**
- **Analytical Thinking**
- **Research Skills**
- **Adaptability**

Detail-Oriented | Critical Thinking | Collaboration | Time Management

LANGUAGES

- **English**
- **Kinyarwanda**

REFERENCES

Prof. Blaise TCHAPNDA
Academic Director
AIMS Rwanda
Email: blaise.tchapnda@aims.ac.rw

- **Tools:** R, Python, Mathematical Modeling, Optimization Techniques
- Developed models to optimize allocation of limited public health resources for cholera prevention interventions
- Evaluated intervention strategies to support evidence-based decision-making in resource-constrained settings
- Applied epidemiological and statistical methods to improve health outcomes and resource efficiency

Rwanda Population Projection 2050

University of Rwanda

2024–2025

- **Tools:** Python, R, Excel, Demographic Analysis Software
- **Findings:** Projected 40% population growth by 2050 with significant aging demographic shift
- Analyzed labor market implications and policy recommendations

Malignant Melanoma Survival Analysis

Advanced Data Analysis

2026

- **Tools:** R (survival analysis packages), Cox Proportional Hazards Model
- Identified ulceration (HR=3.20) and tumor thickness (HR=1.115) as key mortality predictors
- Model achieved 75.3% concordance in distinguishing patient risk

Geometric Jigsaw: Shape Dissections

AIMS Rwanda

2025

- **Tools:** Geometric Software (Geogebra), Mathematical Modeling
- **Findings:** Developed efficient dissection algorithms reducing transformation steps by 25%
- Enhanced spatial reasoning and mathematical problem-solving skills

Strong Password Checker

AIMS Rwanda

2025

- **Tools:** Python, String Manipulation, Logic Implementation
- **Features:** Evaluates password strength based on complexity criteria
- Successfully identified 95% of weak passwords in test datasets

Work-Energy Theorem Application

AIMS Rwanda

2025

- **Tools:** Physics Simulation Software, Mathematical Derivation
- **Findings:** Validated theorem applications in frictionless systems
- Demonstrated energy conservation principles in practical scenarios

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